INORGANIC CUMULATIVE EXAMINATION

March 31, 1990

1. Air oxidation of Co(II) carbonate and aqueous NH_4Cl gives a pink chloride salt with a ratio of $4NH_3$:Co. On addition of HCl to a solution of this salt, a gas is rapidly evolved and the solution slowly turns violet on heating. Complete evaporation of the violet solution yields $CoCl_3 \cdot 4NH_3$. When this is heated in concentrated HCl, a green salt can be isolated which analyzes as $CoCl_3 \cdot 4NH_3 \cdot HCl$. Write balanced equations for all the transformations occurring after the air oxidation. Give as much information as possible concerning the isomerism occurring and give the basis of your reasoning. If you know that the form of $[CoCl_2(en)_2]^+$ that is resolvable into enantiomers is violet, is that helpful?

2. When Co(II) salts are oxidized by air in a solution containing NH_3 and $NaNO_2$, a yellow solid, $[Co(NO_2)_3(NH_3)_3]$, can be isolated. In solution it displays a conductivity small enough to be attributed to impurities only. Werner treated it with HCl to give a complex which, after a series of further reactions, he identified as *trans*-$[CoCl_2(NH_3)_3(OH_2)]^+$. It required an entirely different route to prepare *cis*-$[CoCl_2(NH_3)_3(OH_2)]^+$. Is the yellow substance *fac* or *mer*? What assumption must you make in order to arrive at a conclusion?

3. Why is it not possible to separate and isolate the isomers *cis*-$[CuCl_2(NH_3)_4]$ and *trans*-$[CuCl_2)NH_3)_4]$. Is it likely to be possible to separate and isolate the isomers *cis*-$[IrCl_2(NH_3)_4]^+$ and *trans*-$[IrCl_2(NH_3)_4]^+$?